

Structural Break and Time Series Modeling of Weekly Cereal Sales

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Introduction

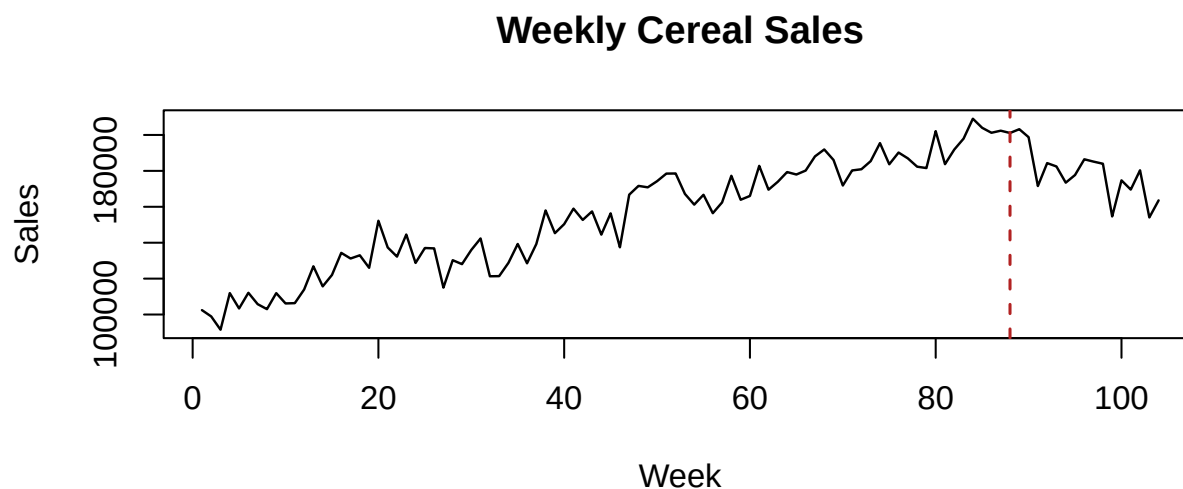
This report examines a published example containing 104 weeks of cereal sales. In the scenario accompanying the data, a competing product enters the market in week 88. The analysis asks whether the sales level or weekly trend changed at that point while accounting for serial correlation.

The dataset comes from the weekly cereal-sales intervention example in Montgomery, Jennings, and Kulahci's *Introduction to Time Series Analysis and Forecasting*, Second Edition, and is available through Wiley's companion materials. The source does not identify the brands or establish whether the values are observed or simulated.

The analysis has three goals:

1. Estimate the immediate level change and the change in trend at week 88.
2. Model autocorrelation rather than relying on ordinary least-squares inference.
3. Produce short-term model-based forecasts and describe their limitations.

Data and Exploratory Analysis



Sales rise through most of the series and then decline after week 88. A segmented trend is therefore a natural description of the visible change. The event timing is supplied with the example, so the analysis treats week 88 as known rather than searching the series for a break point.

The design is an interrupted time series without a control series or additional covariates. It estimates a change in the brand's weekly sales associated with the stated event date, but it cannot establish that competitor entry caused the change; market share is not observed.

Segmented Regression

Let I_t equal one beginning in week 88, and let $A_t = \max(0, t - 88)$. The deterministic part of the model is

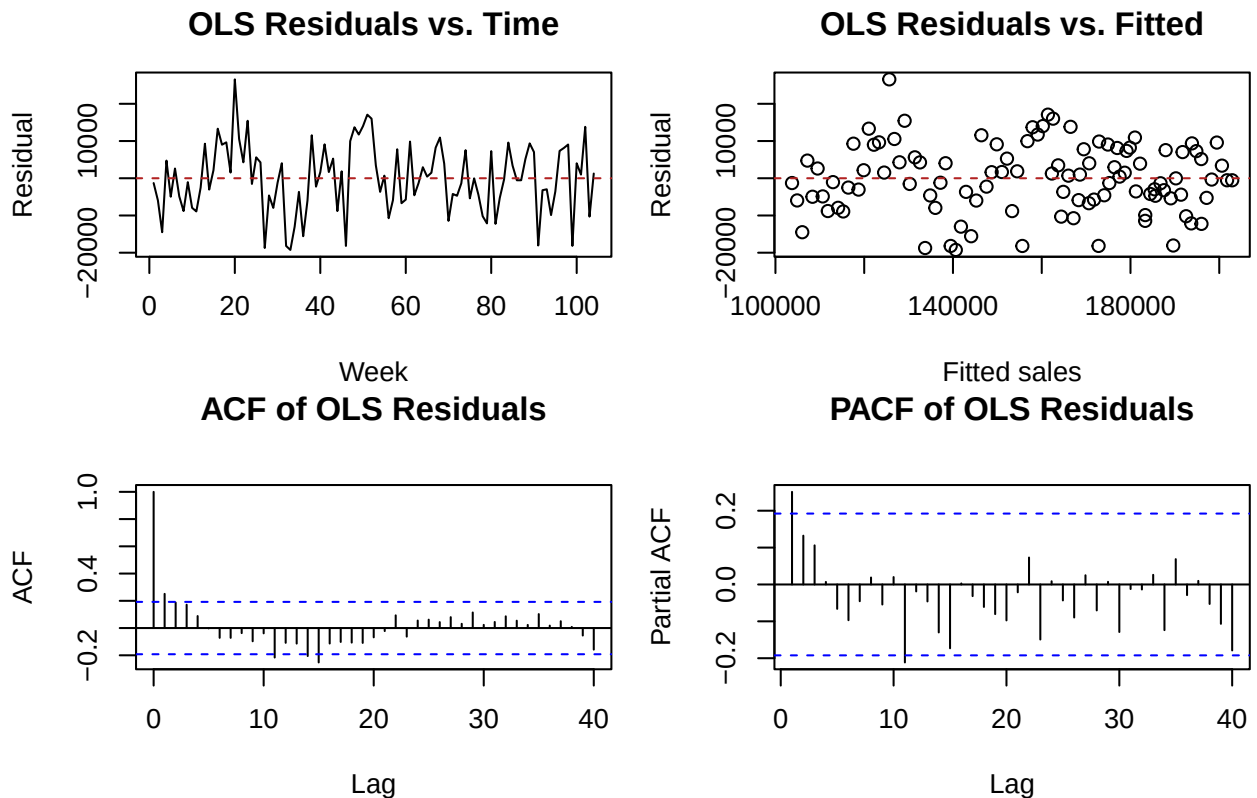
$$y_t = \beta_0 + \beta_1 t + \beta_2 I_t + \beta_3 A_t + e_t.$$

This centered parameterization makes the quantities of interest explicit:

- β_1 is the pre-entry weekly trend.
- β_2 is the immediate level shift at week 88 relative to the pre-entry trend.
- β_3 is the change in weekly trend.
- $\beta_1 + \beta_3$ is the post-entry weekly trend.

An ordinary least-squares fit is useful for inspecting the deterministic structure, but its standard errors are not used for final inference if the residuals are autocorrelated.

Diagnostic	p-value
Ramsey RESET	0.521
Durbin-Watson (positive autocorrelation)	0.002
Nonconstant variance score test	0.291
Shapiro-Wilk	0.514



The Durbin-Watson result and the lag-one autocorrelation show that independence is not a reasonable OLS assumption. The other tests and plots do not flag strong nonlinearity, nonconstant variance, or nonnormality, but those p-values are exploratory because they were computed from dependent residuals. The KPSS test does not reject level stationarity ($p > 0.1$), which supports using a stationary error process without differencing; it does not prove stationarity.

Only two annual cycles are observed. The displayed ACF and PACF cover lags through 40, so annual seasonality at lag 52 cannot be assessed reliably from this sample. No seasonal terms are included.

Joint Regression with Correlated Errors

The regression coefficients and error process are next estimated jointly by maximum likelihood. AR(1), MA(1), and ARMA(1,1) errors are compared because the OLS residual ACF and PACF concentrate most of their structure at short lags.

Error model	AIC	BIC	Ljung-Box p-value (lag 10)
AR(1)	2196.0	2211.9	0.831
MA(1)	2197.4	2213.3	0.619
ARMA(1,1)	2195.6	2214.1	0.950

ARMA(1,1) has the smallest AIC, but it improves on AR(1) by less than one AIC point; BIC instead favors the simpler AR(1). Both specifications leave no evident short-lag autocorrelation. Because the additional moving-average term does not materially improve fit or diagnostics, AR(1) is retained as the primary specification and ARMA(1,1) as a sensitivity check.

For the selected specification,

$$y_t = \beta_0 + \beta_1 t + \beta_2 I_t + \beta_3 A_t + e_t,$$

$$e_t = \phi e_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2).$$

Quantity	Estimate	95% lower	95% upper	p-value
Pre-entry weekly trend	1,152	1,055	1,248	<0.001
Immediate level shift at week 88	-7,552	-18,586	3,482	0.180
Change in weekly trend	-3,294	-4,367	-2,222	<0.001
Post-entry weekly trend	-2,143	-3,208	-1,077	<0.001

Substituting the rounded maximum-likelihood estimates gives

$$y_t = 102,617 + 1,152t - 7,552I_t - 3,294A_t + e_t,$$

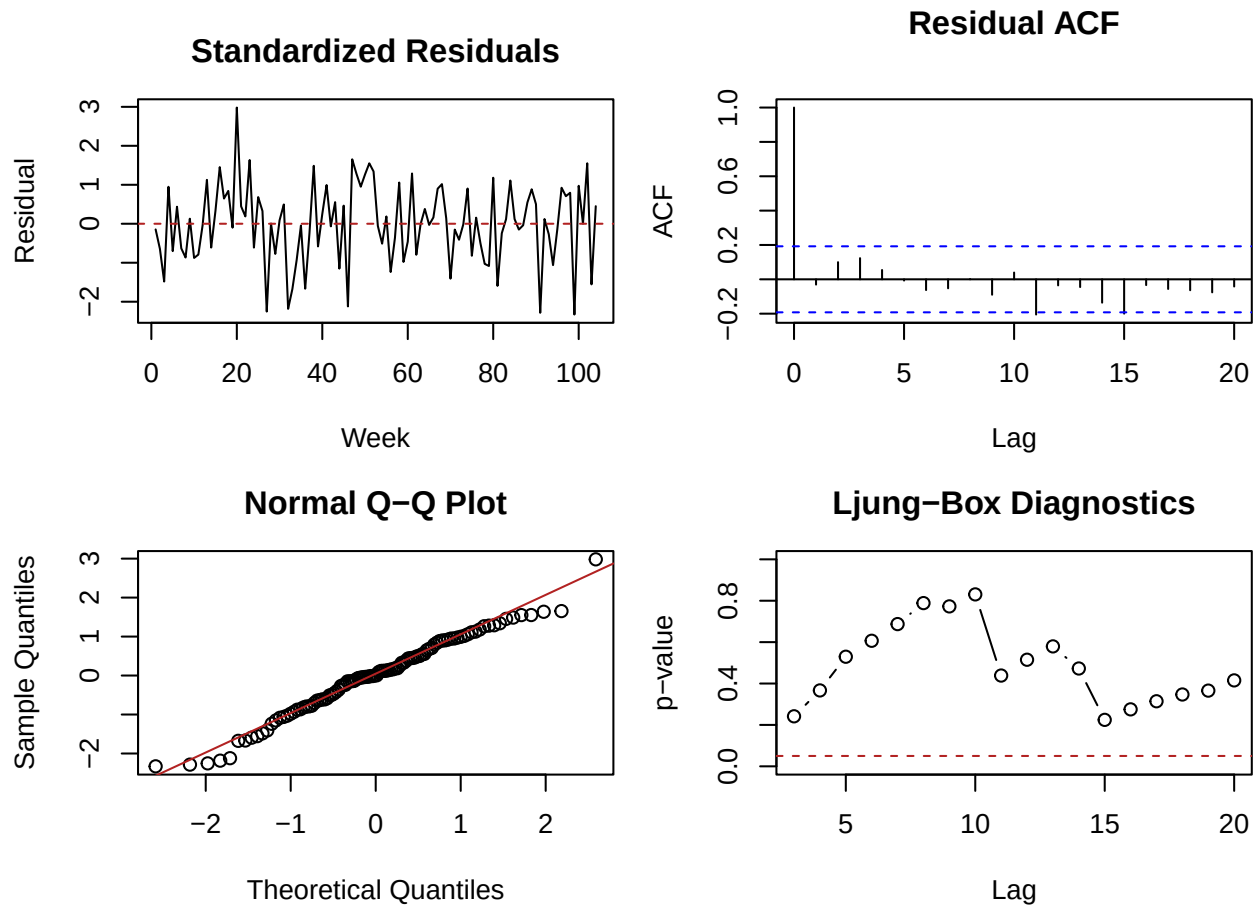
$$e_t = 0.249e_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \hat{\sigma}^2).$$

The fitted pre-entry trend is approximately 1,152 units per week. The estimated post-entry trend is approximately -2,143 units per week, a change of -3,294 units per week. A likelihood-ratio comparison against an AR(1) trend-only model supports adding the level and slope changes ($p < 0.001$). The fitted AR(1) error coefficient is 0.249 ($p = 0.008$).

The immediate level-shift interval includes zero ($p = 0.180$), so the data do not show a statistically clear jump in week 88. The evidence is concentrated in the change in trend. The ARMA(1,1) sensitivity fit produces the same substantive conclusion: an estimated post-entry trend of -2,237 units per week and no statistically clear immediate shift ($p = 0.282$).

The following checks are applied to the innovations from the selected AR(1) model rather than to the initial OLS residuals.

Selected-model diagnostic	p-value
Ljung-Box residual autocorrelation (lag 10)	0.831
Shapiro-Wilk normality	0.260
ARCH LM conditional heteroskedasticity (5 lags)	0.841



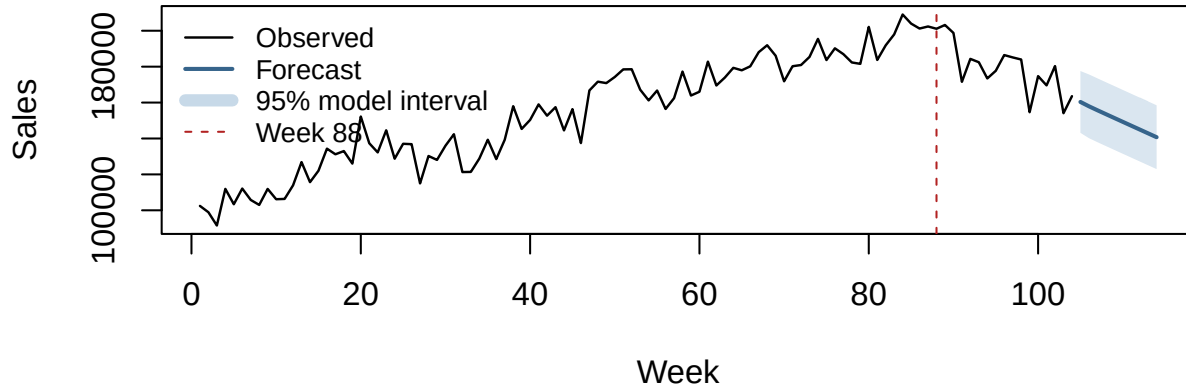
The lag-10 Ljung-Box test does not reject the null of no residual autocorrelation through lag 10 ($p = 0.831$). The Shapiro-Wilk test and Q-Q plot do not flag strong nonnormality ($p = 0.260$), and the five-lag ARCH test does not flag conditional heteroskedasticity ($p = 0.841$). Together with the residual plots, these results support treating the fitted innovations as approximately white noise for this analysis; they do not prove independence, normality, or constant variance. The fitted AR root lies outside the unit circle, satisfying the stationarity condition.

Forecasting

The fitted model is projected ten weeks beyond the observed series. These are conditional, model-based forecasts: they assume the segmented post-entry trend and AR(1) error structure continue unchanged.

Week	Forecast	95% lower	95% upper
105	160,349	143,133	177,566
106	157,940	140,197	175,684
107	155,731	137,956	173,507
108	153,572	135,795	171,350
109	151,426	133,648	169,203
110	149,282	131,504	167,059
111	147,139	129,362	164,917
112	144,996	127,219	162,774
113	142,854	125,076	160,631
114	140,711	122,934	158,489

Observed Sales and 10-Week Model Projection



As a limited forecast check, the AR(1) and ARMA(1,1) specifications were refit at ten rolling origins to predict weeks 95 through 104 one step ahead.

Measure	Result
AR(1) regression RMSE	13,098
ARMA(1,1) regression RMSE	13,013
Naive RMSE	15,107
AR(1) 95% interval coverage	9 of 10 forecasts
ARMA(1,1) 95% interval coverage	8 of 10 forecasts

Both regression models have lower one-step RMSE than the naive benchmark in this short window. The ARMA(1,1) RMSE is only 85 units (0.6%) lower than the AR(1) RMSE, while AR(1) covers 9 of 10 nominal 95% intervals versus 8 for ARMA(1,1). Only 10 forecasts are evaluated, with 87 pre-entry plus 7 post-entry observations available at the first origin, so the comparison does not establish a meaningful predictive advantage for either error model. The ten-week projection should therefore be read as an illustration of the selected AR(1) model rather than a validated long-range forecast.

Conclusion

Jointly estimating the segmented regression and AR(1) errors yields a clear change in weekly trend after week 88. Sales rise by approximately 1,152 units per week before the stated entry date and decline by approximately 2,143 units per week afterward. The immediate level-shift estimate is uncertain and does not distinguish a jump from zero.

The ARMA(1,1) sensitivity fit produces a similar trend reversal, so the substantive finding is not tied to the selected low-order error structure. Residual diagnostics do not flag remaining short-lag autocorrelation, strong nonnormality, or conditional heteroskedasticity, while the limited forecast check shows modest one-step predictive value and does not distinguish the two error models decisively. Because this is an uncontrolled textbook example with a short post-entry period, the analysis supports an association between the event timing and the brand's sales trend, not a causal claim; market share is not observed.

Data Source

Montgomery, D. C., Jennings, C. L., and Kulahci, M. (2015). *Introduction to Time Series Analysis and Forecasting* (2nd ed.). Wiley. Weekly cereal-sales intervention example and [companion data](#).